DTC P0191:00 [SKYACTIV-D 1.5]

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DTC P0191:00	Fuel pressure sensor circuit range/performance problem
DETECTION CONDITION	• The difference between the maximum and minimum value of the fuel pressure sensor output voltage in **less than 0.0025 V** or the difference between the actual fuel pressure and target fuel pressure relative to the feedback **exceeds 5 MPa {51 kgf/cm^2, 725 psi}** for a continuous **1 s** when the following conditions are met. **MONITORING CONDITIONS** — Engine speed: **50—5,200 rpm** — Fuel temperature: **90 °C {194 °F} or less** — Target fuel pressure: **more than 20 MPa {204 kgf/cm^2, 2,901 psi}** — Target fuel injection amount: **exceeds 5 mm^3** — Target fuel discharge amount from supply pump: **less than 1,000 mm^3/rev** — The following DTCs are not detected: • Fuel pressure sensor: P0192:00, P0193:00 • Fuel temperature sensor: P0182:00, P0183:00 • Suction control valve: P062A:00 • Fuel leakage detected: P0093:00 **Diagnostic support note** • This is an intermittent monitor (fuel system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the EGR control. • PCM restricts engine-transaxle integration control. (ATX) • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Fuel pressure sensor connector or terminals malfunction • PCM connector or terminals malfunction • Fuel pressure sensor malfunction • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD FREEZE FRAME DATA/SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (fuel system related) on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT FUEL PRESSURE SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the fuel pressure sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 6.
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 6.
		No	Go to the next step.
5	**INSPECT FUEL PRESSURE SENSOR** • Reconnect all disconnected connectors. • Inspect the fuel pressure sensor. (See FUEL PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the common rail, then go to the next step. (See COMMON RAIL REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
6	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Drive the vehicle on a flat road surface 5 times under the following conditions. 1. Idle the engine for 20 s. 2. Shift to 2nd gear and rapidly accelerate the vehicle to **35 km/h {22 mph}** and drive the vehicle at a constant speed for **25 s**. 3. Stop the vehicle. 4. Switch the ignition to off and wait for **20 s or more**. 5. Shift to 3rd gear and rapidly accelerate the vehicle to **50 km/h {32 mph}** and drive the vehicle at a constant speed for **50 s**. 6. Idle the engine for **20 s**. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
7	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.